or poorly documented. In these scenarios, relying solely on `unknown` may not be feasible due to the complexity of validation and the risk of unsafe property access. Instead, consider combining partial validation with type assertions for critical fields, leveraging TypeScript utility types for dynamic parts, and using schema validation libraries to automate and simplify type checking. Clearly documenting these decisions and establishing team guidelines helps ensure code quality and maintainability even when strict type safety is challenging to enforce.

The `unknown` type is used to represent values of uncertain type while maintaining type safety. The `any` type is used only when it is necessary to bypass TypeScript's type system, but this sacrifices safety and can lead to runtime errors. The best practice is to always prefer `unknown` over `any` when the type of a value is not known, particularly for data from external sources such as API responses, user input, or parsed JSON.

For this topic, are you able to answer these follow-up/probing questions?

- "How would you refactor legacy code that uses `any` to improve type safety, and what role does `unknown` play in this process?"
- "Can you explain a scenario where using `unknown` would be insufficient or impractical, and what alternatives would you consider?"
- "What are the consequences of accidentally assigning an `unknown` value to a variable without proper type checking?"
- "How would you handle a scenario where a function must process values that could be of `any` type, but you want to minimize runtime errors and maximize type safety?"
- "How does the `never` type fit into the landscape of `any` and `unknown`, and in what scenarios would you use it?"

3.8 Functions, classes, and advanced types

Mastering TypeScript's handling of functions, classes, and advanced type features is essential for building scalable and maintainable applications. In technical interviews, candidates are evaluated not only on their ability to write syntactically correct code but also on how effectively they can apply these features to solve real-world architectural and design challenges.